

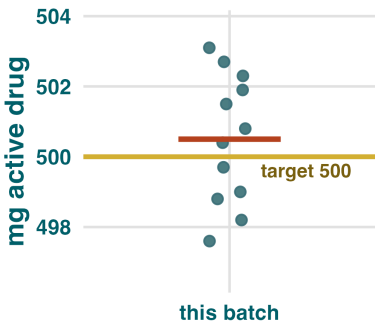
The t-test family | each type, its hypotheses, and its direction

ONE-SAMPLE

One group's mean vs a **fixed target**.
e.g. is a batch's mean dose really the labelled 500 mg?



?
really 500?



Gold line: 500 mg target. Red bar: batch mean.

```
pill <- c(498.2, 501.5, 499.7, 502.3, 500.8,
         497.6, 503.1, 499.0, 501.9, 500.4,
         498.8, 502.7)
t.test(pill, mu = 500) # alpha = 0.05
#> t = 0.94, df = 11, p-value = 0.368
```

R

HYPOTHESES & DIRECTION

$H_0 : \mu = 500$

two-sided $H_1 : \mu \neq 500$ `t.test(pill, mu = 500, alternative = "two.sided")`

one-sided $H_1 : \mu > 500$ `t.test(pill, mu = 500, alternative = "greater")`

one-sided $H_1 : \mu < 500$ `t.test(pill, mu = 500, alternative = "less")`

Here: two-sided. Over- or under-dosing both fail, so watch both tails. The cautious default.

A nurse compares the resting heart rate of 20 athletes with the adult norm of 72 bpm, looking for any difference. Which alternative hypothesis fits?

✓ C

A. $\mu > 72$

B. $\mu < 72$

C. $\mu \neq 72$

For the athlete heart-rate test against the adult norm of 72 bpm, what is the null hypothesis?

✓ B

A. $\mu > 72$

B. $\mu = 72$

C. $\mu < 72$

Her test returns $p = 0.02$ at $\alpha = 0.05$. What should she conclude?

✓ A

A. Reject H_0

B. Fail to reject H_0

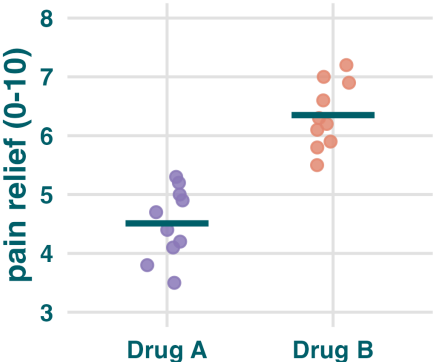
C. H_0 is proven

TWO-SAMPLE (INDEPENDENT)

Means of **two separate groups**.
e.g. does painkiller A or B give more relief?



vs
more relief?



Two independent clouds. Teal bars: group means.

```
A <- c(4.1, 5.2, 3.8, 4.9, 4.4,
       5.0, 3.5, 4.7, 4.2, 5.3)
B <- c(6.3, 5.8, 6.9, 6.1, 7.2,
       5.5, 6.6, 7.0, 6.2, 5.9)
t.test(B, A) # alpha = 0.05
#> t = 7.04, df = 17.9, p-value < 0.001
#> p < 0.05 -> reject H0
```

R

HYPOTHESES & DIRECTION

$H_0 : \mu_A = \mu_B$

two-sided $H_1 : \mu_A \neq \mu_B$ `t.test(B, A, alternative = "two.sided")`

one-sided $H_1 : \mu_B > \mu_A$ `t.test(B, A, alternative = "greater")`

one-sided $H_1 : \mu_B < \mu_A$ `t.test(B, A, alternative = "less")`

Either can apply. Use one-sided only if "is B better?" is the whole question, decided **before** the data.

Researchers compare mean cholesterol in two separate groups: one on a Mediterranean diet, one on their usual diet. What is the null hypothesis?

✓ A

A. $\mu_{\text{med}} = \mu_{\text{usual}}$

B. $\mu_{\text{med}} < \mu_{\text{usual}}$

C. $\mu_{\text{med}} \neq \mu_{\text{usual}}$

Why are two-sided t-tests often preferred in clinical drug testing?

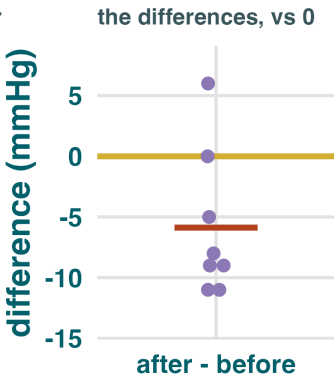
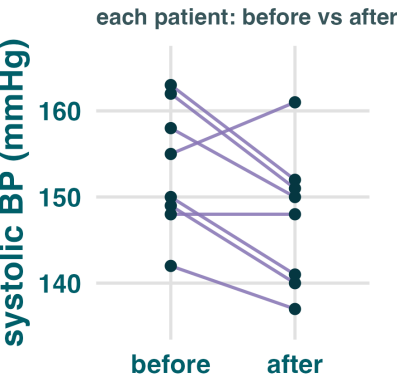
✓ B

A. They make it easier to claim a drug works by lowering the p-value.

B. They detect either improvement or underperformance, and reduce directional bias.

PAIRED (DEPENDENT)

The **same group** measured **twice**.
e.g. does BP fall from before to after a drug?



Each before→after pair becomes one difference, tested against 0.

```
before <- c(150,148,162,142,155,149,163,158)
after <- c(141,148,151,137,161,140,152,150)
t.test(after, before, paired = TRUE) # a=0.05
t.test(after - before, mu = 0) # same
#> t = -2.76, df = 7, p-value = 0.028
#> p < 0.05 -> reject H0
```

R

HYPOTHESES & DIRECTION ($d = \text{after} - \text{before}$)

$H_0 : \mu_d = 0$

two-sided $H_1 : \mu_d \neq 0$ `t.test(after, before, paired = TRUE, alternative = "two.sided")`

one-sided $H_1 : \mu_d < 0$ `t.test(after, before, paired = TRUE, alternative = "less")`

one-sided $H_1 : \mu_d > 0$ `t.test(after, before, paired = TRUE, alternative = "greater")`

For each study, name the t-test and a fitting H_1 :

1. Same runners timed before and after an 8-week program, hoping for faster times.

✓ paired, $H_1 : \mu_d < 0$

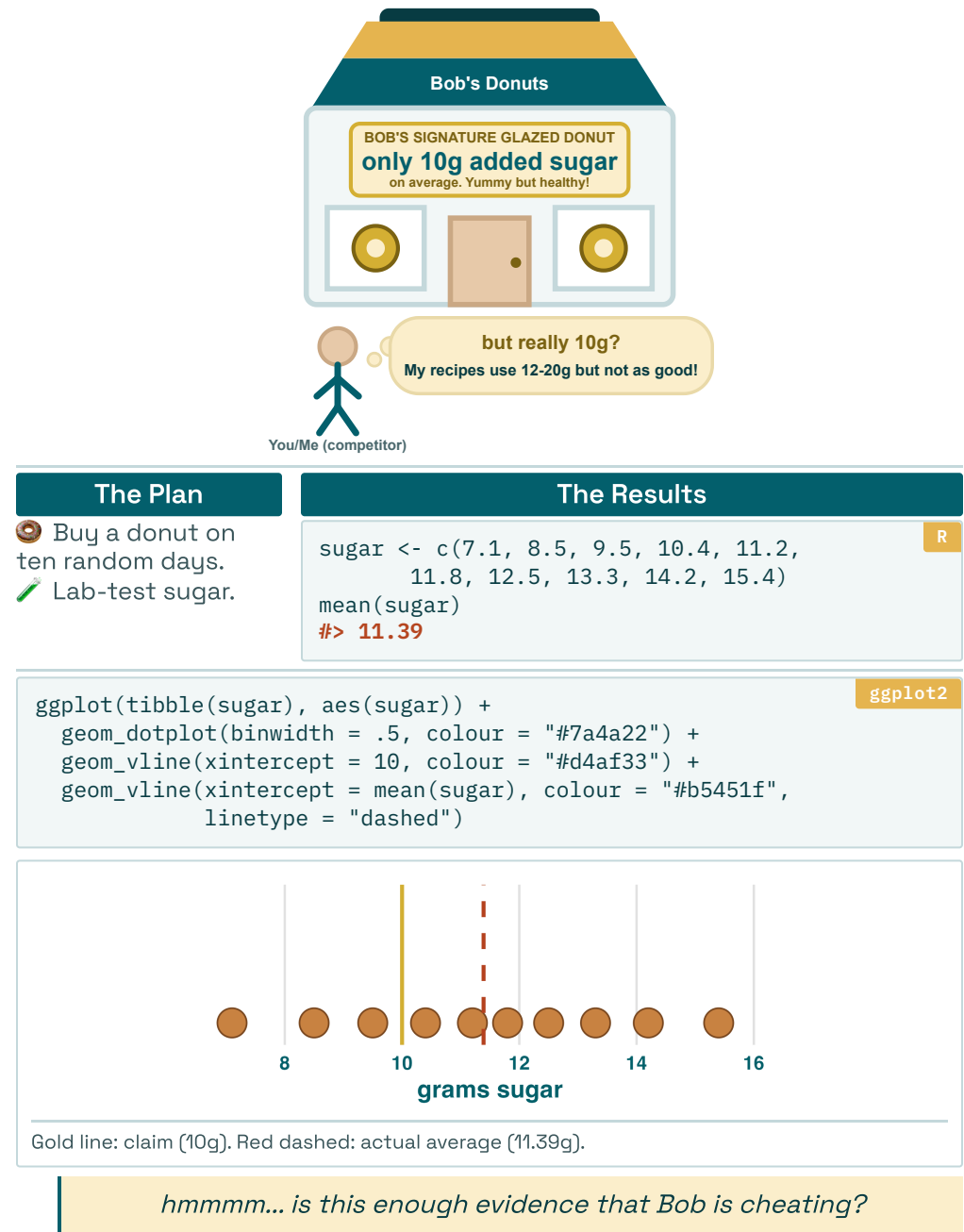
2. Mean recovery time for two surgical methods, on two separate groups of patients.

✓ two-sample, $H_1 : \mu_1 \neq \mu_2$

3. Mean volume in a sample of vials, checked against the labelled 0.5 mL.

✓ one-sample, $H_1 : \mu \neq 0.5$

Bob's Donut Cafe | Introduction to the one-sample t-test



Practice reading one-sample t-tests, then extend Bob's audit

The built-in datasets `sleep` data record extra hours of sleep after two drugs. Here we test whether drug 2 adds more than 0 hours on average.

R

```
data(sleep, package = "datasets")
drug2 <- subset(sleep, group == 2)$extra
drug2
# [1] 1.9 0.8 1.1 0.1 -0.1 4.4 5.5
1.6 4.6 3.4

t.test(drug2, mu = 0,
       alternative = "greater")
```

One Sample t-test

data: drug2

t = 3.6799, df = 9, p-value = 0.002538

alternative hypothesis: true mean is greater than 0

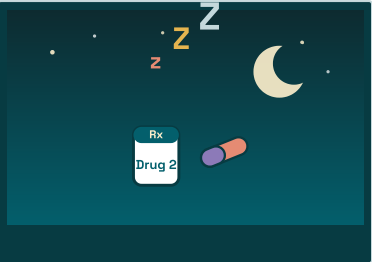
95 percent confidence interval:

1.169334 Inf

sample estimates:

mean of x

2.33



1. What is the null hypothesis being tested?

✓ A

A. $H_0 : \mu = 0$

B. $H_0 : \mu > 0$

C. $H_0 : \mu = 2.33$

2. What is the sample mean extra sleep? ✓ B

A. 0

B. 2.33

C. 3.6799

3. How many observations were tested? ✓ C

A. 9

B. 2

C. 10

4. Match each term to its value, then to its plain-English meaning. Some values are rounded.

Values: a. 9 · b. 0.002538 · c. [1.17, ∞) · d. 3.68

TERM	VALUE	MEANING
t	d. 3.68	b. ≈ # of SDs sample mean is above 0
df	a. 9	a. sample size minus one (n - 1)
p-value	b. 0.002538	d. P(t this extreme H_0 true)
95% CI	c. [1.17, ∞)	c. plausible range for the true mean μ

Meanings: a. n - 1 · b. ≈ # of SDs the sample mean sits above 0 · c. plausible range for μ · d. P(t this extreme | H_0)

5. At $\alpha = 0.05$, what is the decision? ✓ B

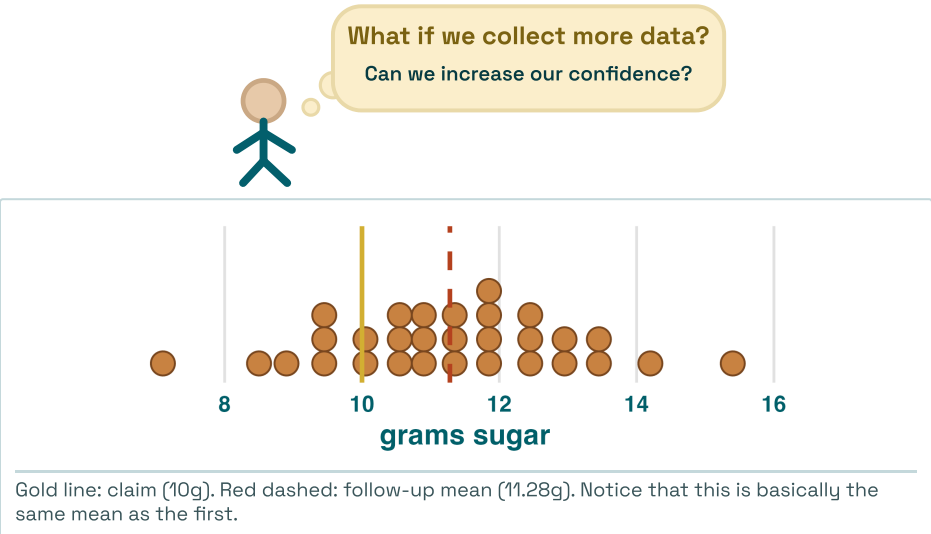
A. Fail to reject H_0

B. Reject H_0

C. H_0 is proven true

Back to the audit of Bob's donuts: More data

The first donut audit had $\bar{x} = 11.39$ and one-sided $p = 0.062$. Close, but not below $\alpha = 0.05$.



```
sugar30 <- c(7.1, 8.5, 9.5, 10.4, 11.2,
11.8, 12.5, 13.3, 14.2, 15.4,
10.9, 9.6, 11.7, 12.3, 10.2,
11.0, 13.1, 9.9, 12.6, 10.7,
11.5, 10.4, 12.0, 9.3, 11.9,
13.6, 10.8, 11.3, 12.8, 8.9)
```

```
t.test(sugar30, mu = 10,
       alternative = "greater")
```

One Sample t-test

data: sugar30

t = 3.9266, df = 29, p-value = 0.000244

alternative hypothesis: true mean is greater than 10

95 percent confidence interval:

10.72612 Inf

sample estimates:

mean of x

11.28

Manual calculations

$s = 1.79 \text{ g}$

$SE = \frac{s}{\sqrt{n}} = \frac{1.79}{\sqrt{30}} = 0.33$

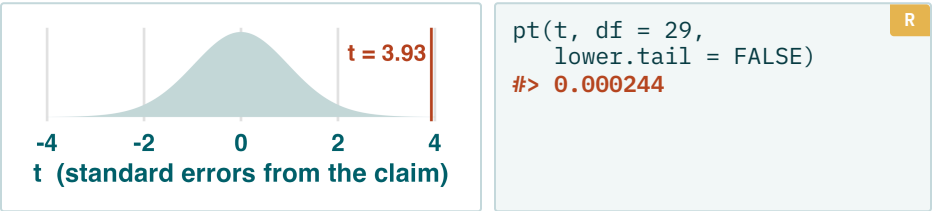
$t = \frac{11.28 - 10}{0.33} = 3.93$

```
s <- sd(sugar30)
#> 1.79

se <- s / sqrt(30)
#> 0.33

t <- (mean(sugar30) - 10) / se
#> 3.93
```

Convert t to p



Interpret the follow-up test

6. What stayed the same from the first donut test? ✓ A

A. $H_0 : \mu = 10$ and $H_1 : \mu > 10$

B. The sample size

C. The p-value

7. Why is the evidence now stronger? ✓ C

A. The claim changed

B. The sample mean became significantly larger

C. More data made the SE smaller

8. At $\alpha = 0.05$, what is the decision now? ✓ B

A. Fail to reject H_0

B. Reject H_0

C. Prove H_0 false

9. In context, the follow-up audit gives strong evidence that Bob's true mean added sugar is ____ 10g. (above / below / equal to) ✓ above

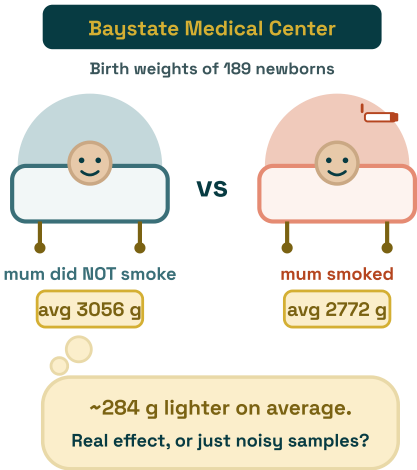
10. Which interpretation is most careful? ✓ C

A. Every donut has more than 10g of sugar

B. Bob is probably adding 11.28g of sugar to all his donuts

C. The data are unlikely if Bob is truly adding 10g sugar on average

Intro to the two-sample t-test



The Data

🏥 189 births at one US hospital.

⚖️ Weigh each newborn.

👤 Split by whether mum smoked.

Group means

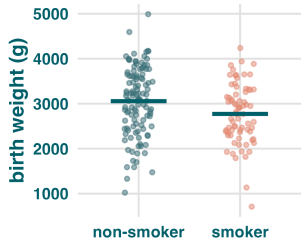
```
library(dplyr)
data(birthwt, package = "MASS")

birthwt <- birthwt |>
  mutate(smoke = factor(smoke,
    labels = c("non-smoker", "smoker")))

birthwt |>
  group_by(smoke) |>
  summarise(mean_bwt = mean(bwt))

#> smoke      mean_bwt
#> non-smoker    3056
#> smoker       2772
```

```
ggplot(birthwt, aes(smoke, bwt)) +
  geom_jitter(width = .14, alpha = .5) +
  stat_summary(fun = mean, geom = "crossbar")
```



Two independent groups. Teal bars: group means (3056 g vs 2772 g).

Smokers’ babies look ~284 g lighter. Is that real, or just two noisy samples?

Hypothesis test framework

A two-sample test compares **two group means**. H_0 says they are equal. H_0 is the status quo. H_1 is what the data would need to give evidence for.

NULL HYPOTHESIS

$H_0 : \mu_{\text{non}} = \mu_{\text{smoke}}$
the true group means are equal.

ALTERNATIVE HYPOTHESIS

$H_1 : \mu_{\text{non}} \neq \mu_{\text{smoke}}$
the true group means differ.

Quick check: name the hypotheses

Comparing mean cholesterol in two separate groups (drug vs placebo), what is H_0 ? **✓ B**

A. $\mu_{\text{drug}} < \mu_{\text{placebo}}$
B. $\mu_{\text{drug}} = \mu_{\text{placebo}}$
C. $\mu_{\text{drug}} \neq \mu_{\text{placebo}}$

You want to detect any birth-weight difference, up or down. Which H_1 ? **✓ C**

A. $\mu_{\text{non}} > \mu_{\text{smoke}}$
B. $\mu_{\text{non}} < \mu_{\text{smoke}}$
C. $\mu_{\text{non}} \neq \mu_{\text{smoke}}$

Measuring the gap: is 284 g a lot?

By itself, **284 g is unclear**. We weigh it against how much birth weights vary **within** each group, and how many babies are in **each** group.

Spread within groups

Wide spread → gap unclear

Tight spread → gap clear

Two sample sizes

few babies → gap may be luck

many babies → gap is trustworthy

For the same gap between means, the evidence is stronger when the within-group spread is smaller (smaller / larger) and the group sizes are larger (larger / smaller). **✓ smaller • larger**

Step 1: spread and sample sizes

Mean gap already found: **283.8 g**. For Welch’s test, collect each group’s SD and n .

```
s1 <- sd(filter(birthwt, smoke == "non-smoker")$bwt)
n1 <- sum(birthwt$smoke == "non-smoker")
s2 <- sd(filter(birthwt, smoke == "smoker")$bwt)
n2 <- sum(birthwt$smoke == "smoker")
#> s1 = 752.7, n1 = 115
#> s2 = 659.6, n2 = 74
```

Step 2: standard error of the difference

Combine each group’s wobble. Welch uses separate variances, not a pooled variance.

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{752.7^2}{115} + \frac{659.6^2}{74}} = 104 \text{ g}$$

```
se <- sqrt(s1^2/n1 + s2^2/n2)
#> 104.0
```

Step 3: t statistic

Standardised gap: how many SEs apart are the means?

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{283.8}{104} = 2.73$$

```
t <- 283.8 / se
#> t = 2.73
```

Step 4: p -value

Compare t to a t curve, $df \approx 170$ (Welch). Two-sided H_1 , so p = **both tails**.

```
2 * pt(-abs(t), df = 170)
#> 0.007
```

$t = 2.73$, two-sided $p = 0.007$.

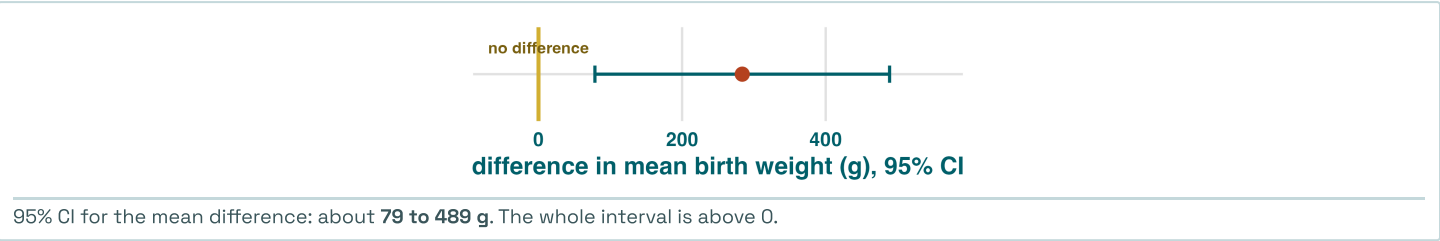
All-in-one in R

```
t.test(bwt ~ smoke, data = birthwt); t.test(birthwt$bwt ~ birthwt$smoke) # same
# Welch Two Sample t-test
data: bwt by smoke
t = 2.7299, df = 170.1, p-value = 0.007003
alternative hypothesis: true difference in means between group non-smoker and group smoker is not equal to 0
95 percent confidence interval:
 78.57486 488.97860
sample estimates:
mean in group non-smoker    mean in group smoker
      3055.696              2771.919
#> p < 0.05 -> reject H0
```


Confidence intervals, p-values, and reading a two-sample t-test

Confidence interval for the birth-weight gap

The estimate was non-smoker mean minus smoker mean. Positive values mean the non-smoker group had heavier babies on average.



	p-value	95% confidence interval
Question	Is 0 plausible?	Which values are plausible?
Reject at .05	$p < .05$	CI excludes 0
Report	$p = .007$	mean diff = 284 g, 95% CI [79, 489]

1. In a small clinic study, patients in a walking programme had mean resting heart rates 4 beats per minute lower than controls, with a 95% CI of [1, 7]. What decision matches at $\alpha = 0.05$?

A. We fail to reject H_0 . B. We reject H_0 . C. We have proven H_0 is false.

2. Reading two CI pictures: which interval gives stronger evidence of a difference?

A. The interval that crosses 0 gives stronger evidence. B. The interval entirely above 0 gives stronger evidence. C. The two intervals give identical evidence.

3. Reading a paper: mean systolic blood pressure was 6 mmHg lower on a new drug, 95% CI [-1, 13], $p = .08$. What is the clean reading?

A. The study gives clear evidence of a drug effect. B. The result is still compatible with no true mean difference. C. The study proves the drug has no effect.

4. Writing a manuscript: "mean difference = 284 g" without a CI is missing what?

A. The sentence is missing precision around the estimate. B. The sentence is missing the group labels only. C. The sentence is missing the sample mean.

- Answers 1-4**
- B. The confidence interval excludes 0, so we reject H_0 at .05.
 - B. An interval entirely above 0 means every plausible value is a positive difference.
 - B. The confidence interval includes 0 and $p > .05$, so the result is compatible with no true mean difference.
 - A. A confidence interval shows precision by showing how uncertain the 284 g estimate is.

ToothGrowth data

The built-in datasets `ToothGrowth` records tooth length in two separate supplement groups: 30 guinea pigs got orange juice (OJ), 30 got ascorbic acid (VC). This is not paired data.

```
t.test(len ~ supp, data = ToothGrowth); t.test(ToothGrowth$len ~ ToothGrowth$supp) # same
```

Welch Two Sample t-test
data: len by supp
t = 1.9153, **df** = 55.309, **p-value** = 0.06063
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-0.171 7.571
sample estimates:
mean in group OJ mean in group VC
20.66 16.96

vitamin C, two ways

1. What is the null hypothesis for comparing the two separate supplement groups?
A. $\mu_{OJ} = \mu_{VC}$
B. $\mu_{OJ} \neq \mu_{VC}$

2. According to the sample estimates, what is the mean tooth length in the OJ group?

A. The OJ mean is 16.96. B. The OJ mean is 20.66. C. The OJ mean is 1.92.

3. Match each term to its value, then to its plain-English meaning. Some values are rounded.

TERMS	VALUES	MEANINGS
t	a. 55.3	a. # of SEs apart
df	b. 0.061	b. adjusted df
p-value	c. [-0.17, 7.57]	c. plausible range
95% CI	d. 1.92	d. P(t this extreme H_0)

4. At $\alpha = 0.05$, what decision should we make for the ToothGrowth test?

A. We fail to reject H_0 because $p = 0.061$. B. We reject H_0 . C. We have proven H_0 is true.

Answers 1-4

- A. The null says the two true supplement means are equal.
- B. The sample estimates show OJ mean = 20.66.
- A. $p = 0.061 > .05$, so fail to reject H_0 .

Term	Value	Meaning
t	d. 1.92	a. # of SEs apart
df	a. 55.3	b. adjusted df
p-value	b. 0.061	d. P(t this extreme H_0)
95% CI	c. [-0.17, 7.57]	c. plausible range